



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 10

TITLE: Write a Java program to demonstrate the use of abstract classes and abstract methods

Problem Statement

Write a Java program to demonstrate the use of abstract classes and abstract methods

Learning Objectives

To implement abstraction using abstract classes and abstract methods.

Theory

Abstract classes provide partial implementation of a program and cannot be instantiated directly. They may contain both abstract and concrete methods. Abstract methods declare functionality that must be implemented by subclasses. Abstraction hides implementation details while exposing only essential functionality. This improves code flexibility and maintainability in object-oriented programming.

Conclusion:

The Java program was successfully implemented using abstract classes and abstract methods to demonstrate the concept of abstraction. The abstract class provided a common structure, while the subclasses implemented the required functionality according to their specific requirements. The Payment System and Food Order applications showed how abstraction hides implementation details and promotes code reusability, flexibility, and maintainability. This experiment strengthened the understanding of object-oriented programming principles in Java.

Exercise

Create an abstract Payment class and implement Credit Card and UPI payment classes.?

```
abstract class Payment {  
    abstract void pay(double amount);  
}
```

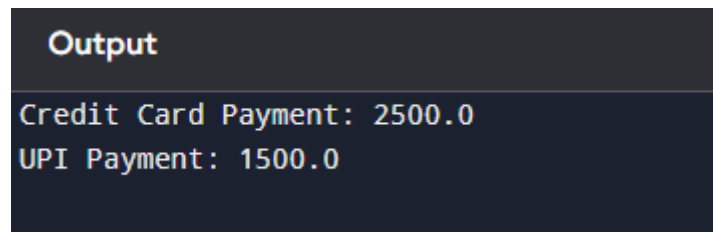
```
class CreditCard extends Payment {  
    void pay(double amount) {  
        System.out.println("Credit Card Payment: " + amount);  
    }  
}
```

```
class UPI extends Payment {  
    void pay(double amount) {  
        System.out.println("UPI Payment: " + amount);  
    }  
}
```

```
class Main {  
    public static void main(String[] args) {  
        Payment p1 = new CreditCard();  
        Payment p2 = new UPI();  
  
        p1.pay(2500);  
    }  
}
```

```
        p2.pay(1500);  
    }  
}
```

OUTPUT



```
Output  
Credit Card Payment: 2500.0  
UPI Payment: 1500.0
```

2. Create an abstract FoodOrder class with an abstract method calculateBill(). Implement two subclasses, DineInOrder and TakeAwayOrder, to calculate and display the total bill based on the order type.

```
abstract class FoodOrder {  
    double amount;  
  
    FoodOrder(double amount) {  
        this.amount = amount;  
    }  
  
    abstract void calculateBill();  
}
```

```
class DineInOrder extends FoodOrder {  
    DineInOrder(double amount) {
```

```
        super(amount);
    }

    void calculateBill() {
        double total = amount + (amount * 0.10);
        System.out.println("Dine-In Bill: " + total);
    }
}

class TakeAwayOrder extends FoodOrder {
    TakeAwayOrder(double amount) {
        super(amount);
    }

    void calculateBill() {
        double total = amount + 30;
        System.out.println("Takeaway Bill: " + total);
    }
}

class Main {
    public static void main(String[] args) {
        FoodOrder f1 = new DineInOrder(500);
        FoodOrder f2 = new TakeAwayOrder(500);
    }
}
```

```
f1.calculateBill();  
f2.calculateBill();  
}  
}
```

OUTPUT

```
Output  
Dine-In Bill: 550.0  
Takeaway Bill: 530.0
```